

Scalable Infrastructure Architecture

Course Code:

Lecture : 4 hr. per week
Tutorial : 0 hr. per week
Practical : 0 hr. per week

Year:
Part:

Course Objective:

The objective of this course is to familiarize the students with knowledge, skills and tools that are essential for setting up the scalable computing resources through multiple frameworks and other associated architectures. It will describe the software components and tools from which scalable infrastructure and resource configuration are constructed. Methods for building scalable cloud applications will be described and explained. At the end of the course, students will be able to design and build scalable systems and turn into as a solution architect.

Course Contents:

1. Introduction

[6 Hrs.]

1.1 Scalable Computing over the Internet

The Age of Internet Computing; Scalable Computing Trends and New Paradigms
The Internet of Things and Cyber-Physical Systems

1.2 Technologies for Network based systems

Multicore CPUs and Multithreading Technologies; GPU Computing to Exascale and Beyond; Memory, Storage, and Wide-Area Networking; Virtual Machines and Virtualization Middleware

2. System Models for Computing

[6 Hrs.]

2.1 Clusters of Cooperative Computers

2.2 Grid Computing Infrastructures

2.3 Peer-to-Peer Network Families

2.4 Cloud Computing over the Internet

Properties, Characteristics, Benefits; Cloud types; private, public and hybrid cloud; Service models IaaS, PaaS, SaaS.

3. Software Models and Deployment

[14 Hrs.]

3.1 Parallel and Distributed Programming Models

3.2 Service-Oriented Architecture (SOA)

Services and Service-Oriented Architecture; Message-Oriented Middleware; Portals and Science Gateways; Workflows in Service-Oriented Architecture

3.3 Moving from a monolithic to a microservice architecture

3.4 Microservices

Principles of Microservices ; Building Blocks

3.5 Deployment and Infrastructure

Virtualization, Dockers and Containers ; Containerization vs virtualization; Docker Architecture; Containerization of microservices ; Programming on Amazon AWS and Microsoft Azure, Google App Engine; Deploying container images in public cloud

4. Scalable platforms

[12 Hrs.]

4.1 Compute Services

- Virtual server, Cloud containers, Cloud server less service, Auto Scaling
- 4.2 Storage Services
Object storage, file storage, and block storage, Archive Storage
- 4.3 Databases
SQL, NoSQL and Graph Database and Database Migration
- 5. Network Infrastructure and Security [12 Hrs.]**
- 5.1 Networking and Content Delivery – Cloud Network, CDN, DNS Services, Load Balancing.
- 5.2 Software Defined Networks
- 5.3 Authentication, Authorization and Accounting
- 5.4 Shared Responsibility model
- 5.5 Cloud security and compliance concepts
- 5.6 Interoperability and Standards
- 5.7 Monitoring, Analytics, Migration
- 6. Performance, Security, and Energy Efficiency [4 Hrs.]**
- 6.1 Performance Metrics and Scalability Analysis
- 6.2 Fault Tolerance and System Availability
- 6.3 Network Threats and Data Integrity
- 6.4 Energy Efficiency in Distributed Computing
- 7. Ubiquitous Computing and the Internet of Things [6 Hrs.]**
- 7.1 Trends in Supporting Ubiquitous Computing.
- 7.2 Performance of Distributed and Scalable Systems
- 7.3 Enabling Technologies for the Internet of Things
- 7.4 Innovative Applications of the Internet of Things
- 7.5 Online Social and Professional Networking

Practical:

There will be lab exercises to provide students with hands on experiences on using online cloud computing platform to enhance the concept in following topics

1. Implementation of Web Server and dynamic hosting on linux instances & server launching.
2. User Management, Group Management, Resource Management, Account Management
3. Server Side Encryption, Client Side Encryption
4. Implementation of Different types of Instances Reserved, On-demand, Spot, Dedicated, Auto Scaling, Load Balancing
5. Implement and configuration storage services: Glacier storage, RDS (Relational Database Service), dynamo dB and configuring backups
6. Experiments with networking: Virtual Private Cloud, NAT, VPC Peering Content Delivery Network (CDN)
7. Experimentation with Monitoring and messaging systems

Evaluation Schemes:

a. Internal Examination

Type	Weightage
Minor tests	50%
Assignments	50%

b. Final Examination

There will be five units with total of 60 marks. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1 and 6	20
2	2	10
3	3	10
4	4	10
5	5	10
Total		60

References:

1. Kai Hwang, Geoffrey C. Fox and Jack J. Dongarra, “Distributed and cloud computing from Parallel Processing to the Internet of Things”, Morgan Kaufmann, Elsevier – 2012.
2. Tim Mather, Subra Kumaraswamy, and Shahed Latif, “Cloud Security and Privacy An Enterprise perspective on Risk and Compliance” O’Reilly, 2009
3. James Turnbull, The Docker Book: “Containerization is the new virtualization”, James Turnbull, 2014
4. Bruce Powel Douglass, Real-time design patterns: robust scalable architecture for real-time systems. Addison-Wesley Professional, 2003.
5. Barrie Sosinsky, “Cloud Computing Bible” John Wiley & Sons, 2010

Digital Innovation Ecosystem
Course Code:

Lecture : 4 hr. per week
Tutorial : 0 hr. per week
Practical : 0 hr. per week

Year:
Part:

Course Objectives:

The objective of the course is to familiarize the students with knowledge, skills, tools and frameworks that are useful for digital innovation and entrepreneurship. After completing the course, students will gain knowledge and skills necessary to analyze and evaluate current digital ecosystem and commence own innovation journey. At the end, the students will be able to develop remarkable ICT related projects that will help to bridge the innovation gap.

Course Outlines:

- 1. Innovation and ICT centric innovation ecosystem [8 Hrs.]**
 - 1.1 Introduction to innovation (opportunities, strategy and disruption, developing and pitching innovation, building ecosystems, lean innovation, financing, selling and marketing, leading entrepreneurial ventures, and globalization)
 - 1.2 Digital transformation, digital innovation, and entrepreneurship
 - 1.3 Digital innovation ecosystem
 - 1.4 An innovation and its stakeholders
 - 1.5 Ecosystem sustainability
- 2. Ecosystem performance analysis [10 Hrs.]**
 - 2.1 Engines of growth for digital transformation
 - 2.2 Assessing ecosystem performance
 - 2.3 Digital innovation divide
 - 2.4 Job-to-be-done framework
 - 2.5 Closing the digital innovation gap
 - 2.6 Entrepreneurial lifecycle and the innovation journey
 - 2.7 Challenges of ICT-centric innovation ecosystems
- 3. Developing sustainable ecosystems [10 Hrs.]**
 - 3.1 Sustainable ecosystems creation
 - 3.2 Ecosystem governance
 - 3.3 Ecosystem linkages and focused ecosystem
 - 3.4 Ecosystem capacity
 - 3.5 Ecosystem development roadmap
- 4. Digital innovation tools [10 Hrs.]**
 - 4.1 Analysis framework
 - 4.2 Stakeholder engagement tool
 - 4.3 Ecosystem assessment canvas
 - 4.4 Ecosystem maturity map
 - 4.5 Service design tool
 - 4.6 Storytelling canvas

5. Digital transformation and innovation journey

[10 Hrs.]

- 5.1 Digital transformation: determinants of digital transformation and use of ecosystem assessment canvas
- 5.2 An innovation journey: entrepreneurial lifecycle and use of ecosystem maturity map
- 5.3 Strategic roadmap: developing and maturing ecosystems and use of strategic priority matrix
- 5.4 The policy paradox: framework to solve complex problems and policy experimentation
- 5.5 Accelerating digital transformation

6. National digital economy and contemporary issues

[12 Hrs.]

- 6.1 Digital strategies and the digital economy
- 6.2 Aligning digital strategies with national priorities
- 6.3 WSIS and Connect 2030 agenda
- 6.4 ICT related policies
- 6.5 Innovation and entrepreneurship in ICT policies
- 6.6 Digital Nepal framework
- 6.7 Profession and ethical decision making
- 6.8 Intellectual Property Rights (IPR)

Practical/Case Study

1. To impart the practical knowledge case studies will be done.
2. Recent article in the related field will be studied.
3. By using the knowledge and tools students will develop remarkable ICT related projects.

Evaluation Schemes:

a. Internal Examination

Type	Weightage
Minor tests	50%
Assignments	50%

b. Final Examination

There will be five units with total of 60 marks. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1 and 2	12
2	3	12
3	4	12
4	5	12
5	6 and 7	12
Total		60

References

1. Hjalmarsson, Anders, Gustaf Juell-Skielse, and Paul Johannesson. Open Digital Innovation. Springer International Publishing, 2017.

2. Mazzarol, Tim, and Sophie Reboud. Entrepreneurship and innovation. Tilde Publishing, 2017.
3. Bridging the Digital Innovation Divide: A toolkit for developing sustainable ICT-centric ecosystem projects, ITU Publications. 2020.
4. Accelerating Digital Transformation: Good practices for developing, driving and accelerating ICT centric innovation ecosystems in Europe, ITU Publications. 2020.
5. International journals related to digital economy and innovation.